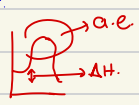
Reaksiyonun ilerlememe Sebebi
Ya aktivasyon enerjisine gelmemiştir
Ya da difüzyon Çok yavaştır.



Kinetik aktivasyon enerjisiyle ilgileniyor.
Günlük hayatta homojen reaksiyon metalurji de yoktur. Fakat temel bilimler (kinetik) homojen reaksiyonlara göre denklemleri oluşturur.

Reaction Rate → (r)

Reaksiyon hızını neye göre hesapladığım önemli.
Girenlerin tükenmesi cinsinden mi? hesaplıyorum.
Yoksa ürünlerin oluşması cinsinden mi?

$aA + bB \longrightarrow cC$

$C = \frac{n}{V}$ $\quad r = -\frac{dc}{dt}$

tükenenden hesaplarsam −

Oluşandan hesaplarsam +

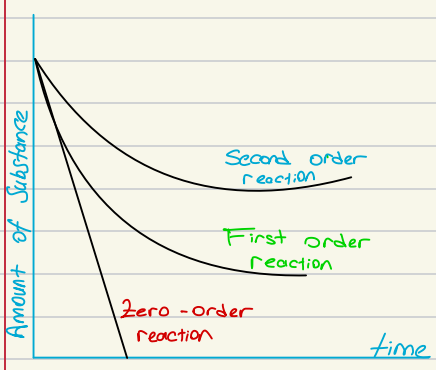
$X = C_0 - C$ $\quad r = \frac{dx}{dt}$

The reaction rate is always possitive.

A negative sign is present to indicate the reactant concentration is decreasing.

Factors Affecting Reaction Rates

Concentration of reactants
Temperature of reactions
Physical State of reactions
Presence of Catalyst or Inhibitor.
Means of excitation.



- Hız Sabiti (k) sadece Sıcaklığa bağlı
- Hız T, C, P 'ye bağlı.

Design of a metalurgical process

The most important questions while designing a metallurgical reactor and process are

1. Which reactions will be realized in the reactor?
2. What will be the rate of these reactions until they reach the equilibrium?

The first question is related with materials thermodynamics while material kinetics is interested with the second one.

Kinetic Analyses of Metallurgical Systems

In addition to the feasibility of a reaction, thermodynamics can predict the compositions of co-existing phases under certain pressure and temperature.

Thermodynamics is interested only in the initial and final states of a transformation process without considering the **reaction path, mechanism of conversion and the time required**.

What path a reaction takes, would again depend over number of details as well as properties of the system that are structure-sensitive. Therefore, **kinetic predictions are mostly either qualitative or quantitative** with some uncertainty.

Kinetic Analyses of Metallurgical Systems

Two questions about a chemical reaction are of paramount importance to engineers engaged in research, design or operation involving chemical reactions:

1. Under what conditions a chemical reaction will be feasible and how far can the reaction proceed.
2. How fast a reaction would take place under some desired operating conditions.

In general, the subject which deals with the measurement of reaction rate is turned as "Reaction Kinetics".

Metalurji Mühendisliği - Kinetik

- Proses içindeki olayları planlayabilmesi, şartları iyileştirebilmesi veya prosese müdehale edebilmesi için **reaksiyon mekanizmasının bilmesi** gereklidir.
- Metalurjide, yüksek basınç ve sıcaklık, zehirli gazlar veya korozif ortam gibi nedenlerle kapalı sistemlerde çalışılır, dolayısıyla olayların gözlemlenebilmesi sınırlıdır ya da mümkün değildir.
- Uygulama esnasında reaktan reaksiyon bitmeden reaktörü terk etmemelidir. **Reaktanların reaktörde kalış süresi, dolayısıyla reaktörün büyüklüğü kinetik deneyler sonucu elde edilen zaman bağıntıları ile planlanır.**
- **Reaksiyon hızlı ise sadece termodinamik veriler yeterlidir.**
- **Reaksiyon hızı düşük ise kinetik veriler de gereklidir.**

Reaction Rate

The reaction rate (rate of reaction) or speed of reaction for a reactant or product in a particular reaction is defined as how fast or slow a reaction takes place. It depends on the concentration (quantitative / extensive properties) of the reactants.

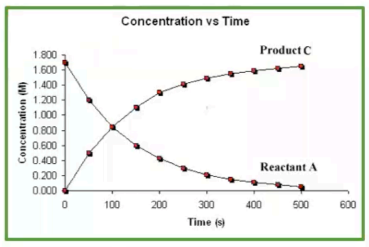
$$aA + bB \rightarrow cC$$

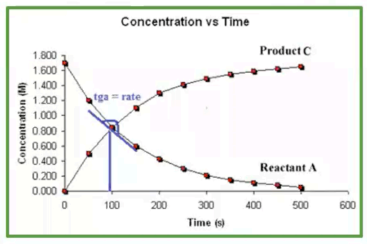
The lowercase letters (*a*, *b*, and *c*) represent stoichiometric coefficients, while the capital letters represent the reactants (A and B) and the product (C).

The rate of a reaction is always positive. A negative sign is present to indicate the reactant concentration is decreasing

$$r = -\frac{dc}{dt} \qquad r = \frac{dx}{dt} \qquad x = C_0 - C \qquad c = \frac{n}{V}$$

Reaction Rate





t = 0	t = t
C_o	C

$$r = -\frac{dc}{dt} \qquad r = \frac{c - c_0}{t - t_0}$$

$c_0 - c = x$

x: reacted part of the reactant / product

Reaction Rate

aA + bB → cC

r = f(T, C, P)
P(constant)

$$r = -\frac{dc_A}{dt} = -\frac{dc_B}{dt} = \frac{dc_C}{dt}$$

k: reaction rate coefficient / rate constant
k = f(T)

$$r_A = -\frac{dc_A}{dt} = kc_A^a c_B^b$$

B + 2D → 3T

$$r_B = -k' c_B c_D^2$$
$$r_D = -k'' c_B c_D^2$$
$$r_T = k''' c_B c_D^2$$

$$k' = \frac{1}{2}k'' = \frac{1}{3}k'''$$

Factors Affecting Reaction Rates

Chemical reaction rates (meaning mass of reactants transformed per unit time) generally depend on the following:

- **Concentrations of reactants** (or more accurately the thermodynamic activities)
- **Temperature of reaction** – increase in temperature increases reaction rate
- **Physical state of reactants** – whether in a gaseous phase or in a solution (homogeneous reactions) or in separate phases (heterogeneous reaction as in a sponge iron rotary kiln) – whether in solid blocks (low surface area) or in powder, colloidal or emulsion form (high surface area). Higher the surface area, higher are the reaction sites and therefore higher are the reaction rates.
- **Presence of Catalyst or Inhibitor**
- **Means of excitation** – Light, Ultraviolet or Radioactive Radiation, etc.

Classification of Reactions

There are several ways of classifying chemical reactions.

- **The basis of mechanism, with respect to the number of molecules actually participation in the reaction, with respect to the operating conditions.**
- **However, the most useful scheme is possibly according to the number and types of phases involved in a chemical reaction.**

Phase	Molecularity	Degree	Mechanism	Catalysis
Homogeneous	One molecular	Zeroth degree	Reversible	Catalytic
Heterogeneous	Two molecular	First degree	Irreversible	Non-catalytic
	Multi molecular	Second degree	Parallel	
		nth degree	Serial	

Classification of Reactions

A reaction is **homogeneous** if it takes place in one phase along. Most of the gas phase reactions are example of homogeneous if it required the presence of two phases to proceed at a rate that it does. Burning of coal, roasting of ores, dissolution of solids in acid fall in the category of **heterogeneous** reaction.

It is to be rated that these classifications are not mutually exclusive. Thus a reaction may be irreversible-second order, Taking place under adiabatic and approximately constant pressure conditions in a continuous type reactor.

They may be **irreversible, reversible, simultaneous or consecutive** type with respect to the number of molecules actually participating in the reaction, it may **unimolecular or bimolecular**. Considering the operating conditions, a chemical reaction maybe **isothermal** at constant pressure, **adiabatic or non-adiabatic** and **non-isothermal**.

Reaksiyonların Sınıflandırılması

Fazlarına göre	Molekülaritesine göre	Derecesine göre	Mekanizmasına göre	Katalizli oluşuna göre
Homojen	Bir moleküllü	Sıfırıncı derece	Dönüşümsüz	Katalitik
Heterojen	İki moleküllü	Birinci derece	Dönüşümlü (tersinir)	Katalitik olmayan
	Çok moleküllü	İkinci derece	Paralel	
		n. derece	Seri	

According to the phases

Homogeneous Reactions: it takes place in one phase along (reactants and products)

$$CO_{(g)} + \tfrac{1}{2} O_{2(g)} = CO_{2(g)}$$

Heterogeneous Reactions: minimum two different phase are required

$$C_{(s)} + O_{2(g)} = CO_{2(g)}$$

All pyrometallurgic reactions, solid-gas, gas-liquid-metal, gas-oxide-liquid reactions are heterogeneous.

S-S, S-L, S-G, L-L, L-G

Order of Reaction

$Ra \rightarrow Rn + He$	$r = k[Ra]^1 = k[Ra]$	1st order
$2HI \rightarrow H_2 + I_2$	$r = k[HI]^2$	2nd order
$2NO + O_2 \rightarrow + 2NO_2$	$r = k[NO]^2[O_2]$	3rd order

According to the order

A rate law is a differential equation that describes **the rate of change of a reactant (or product) concentration with time.**

If we integrate the rate law then we obtain an expression for the concentration as a function of time, which is generally the type of data obtained in an experiment.

Reaksiyon Derecesi

Reaktanların (ya da ürünlerinin) konsantrasyonun değişimini zamana kaçıncı dereceden bağlı olduğudur.

0th konsantrasyona bağlı değilm.

Order of Reaction

The **order of reaction** with respect to a given substance (such as reactant, catalyst or product) is defined as the index, or exponent, to which its concentration term in the rate equation is raised.

A rate law is a differential equation that describes **the rate of change of a reactant (or product) concentration with time**. If we integrate the rate law then we obtain an expression for the concentration as a function of time, which is generally the type of data obtained in an experiment. In many simple cases, the rate law may be integrated analytically.

- For **zero-order reactions**, the reaction rate is independent of the concentration of a reactant, so that changing its concentration has no effect on the speed of the reaction.
- If a reaction rate depends on a single reactant and the value of the exponent is one, then the reaction is said to be **first order**.
- A reaction is said to be **second order** when the overall order is two. The rate of a second-order reaction may be proportional to one concentration squared or (more commonly) to the product of two concentrations

Zeroth Order Reactions

Reaction rate is independent of reactant concentration

$$r = \frac{dx}{dt} = k_0(a-x)^0 = k_0 \qquad r = k_0$$

$$\int dx = \int k_0 dt$$

$$x_0 = k_0 t + cst$$ → constant

"0" ıncı dereceden reaksiyonun hız sabiti.

If $t = 0$, $x = 0$ → $cst = 0$

$$x_0 = k_0 t$$

k_0 = M.sec^{-1}

Zeroth order half life

$$x = k_0 t$$

$$x = \frac{a}{2}$$

$$\frac{a}{2} = k_0 t_{1/2} \qquad t_{1/2} = \frac{a}{2k_0}$$

Half-life is proportional with the concentration

Homogeneous Reaction Kinetics

$\alpha A + \beta B + \gamma C = ...$

$$r = k c_A^{\alpha} c_B^{\beta} c_C^{\gamma}$$

	t = 0	t = t
C_A	a	x
C_B	b	y
C_C	c	z

reaction rate at time t

$$r = k(a-x)^{\alpha}(b-y)^{\beta}(c-z)^{\gamma}$$

If a = b = c and x = y = z

$$r = k(a-x)^{\alpha+\beta+\gamma}$$

$$r = k(a-x)^n$$

$$r = \frac{dx}{dt} = k(a-x)^n$$

reaction rate — rate constant — reaction order

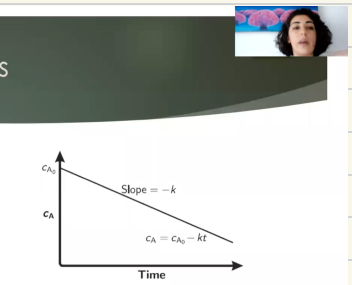
I) Zero Order Reactions (rare)

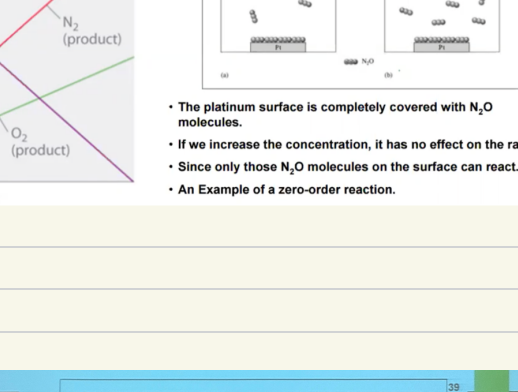
$A \rightarrow$ products

$$-\frac{d[A]}{dt} = k \qquad \{k \text{ is in } [\text{moles}/(\text{liter sec})]\}$$

$$[A] = -kt + [A]_o$$

$t_{1/2} = [A]_o/(2k)$





In Zero Order Reactions,

- **Reaction rate is independent of concentration**
- **Reaction rate vs. time show a lineer dependence**
- **Reaction rate is independent of time**

An example is the decomposition of N_2O on a platinum (Pt) surface to produce N_2 and O_2, which occurs at temperatures ranging from 200°C to 40

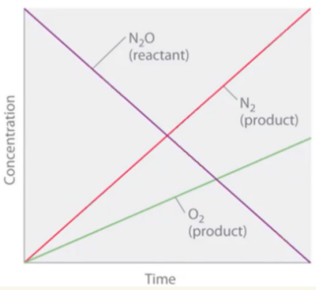
$$2N_2O(g) \xrightarrow{Pt} 2N_2(g) + O_2(g)$$

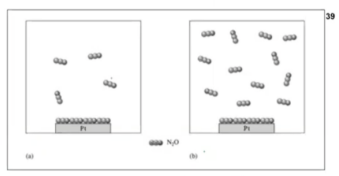
Without a platinum surface, the reaction requires temperatures greater than 700°C, but between 200°C and 400°C, the only factor that determines how rapidly N_2O decomposes is the amount of Pt surface available (not the amount of Pt). As long as there is enough N_2O to react with the entire Pt surface, doubling or quadrupling the N_2O concentration will have no effect on the reaction rate. At very low concentrations of N_2O, where there are not enough molecules present to occupy the entire available Pt surface, the reaction rate is dependent on the N_2O concentration. The reaction rate is as follows:

$$\text{rate} = -\frac{1}{2}\left(\frac{\Delta[N_2O]}{\Delta t}\right) = \frac{1}{2}\left(\frac{\Delta[N_2]}{\Delta t}\right) = \frac{\Delta[O_2]}{\Delta t} = k[N_2O]^0 = k$$

Thus the rate at which N_2O is consumed and the rates at which N_2 and O_2 are produced are independent of concentration. As shown in Figure 14.8, the change in the concentrations of all species with time is linear. Most important, the exponent (0) corresponding to the N_2O concentration in the experimentally derived rate law is not the same as the reactant's stoichiometric coefficient in the balanced chemical equation (2). For this reaction, as for all others, the rate law must be determined experimentally.

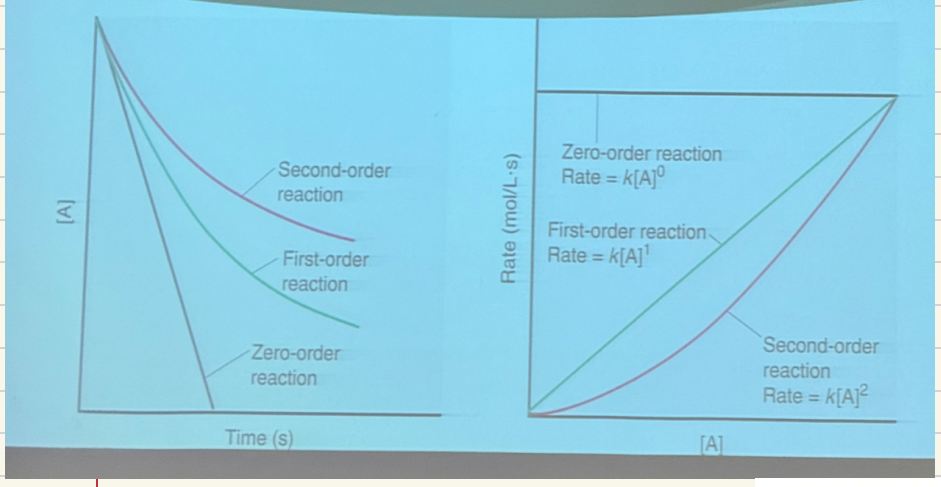
Zeroth order reactions occur most often when the reaction rate is determined by available surface area





- **The platinum surface is completely covered with N_2O molecules.**
- **If we increase the concentration, it has no effect on the rate.**
- **Since only those N_2O molecules on the surface can react.**
- **An Example of a zero-order reaction.**

Order of Reaction



Zeroth Order Reactions

In a zero-order reaction, increasing the concentration of the reactant(s) will not speed up the rate of the reaction.

The most common zero-order reactions are endothermic high-temperature thermal decomposition reactions, where a large amount of thermal energy is required to break a chemical species apart.

In such cases, as long as the temperature is not too high, the available thermal energy controls the rate rather than the concentration of the reactan

Sıfırıncı dereceden bir reaksiyonda, reaktanların konsantrasyonunun arttırılması, reaksiyonun hızını artırmayacaktır. Sıcaklık çok yüksek olmadığı sürece, reaktanların konsantrasyonu değil mevcut termal enerji hızı kontrol eder.

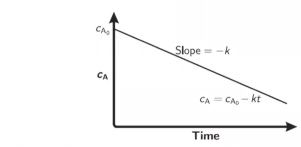
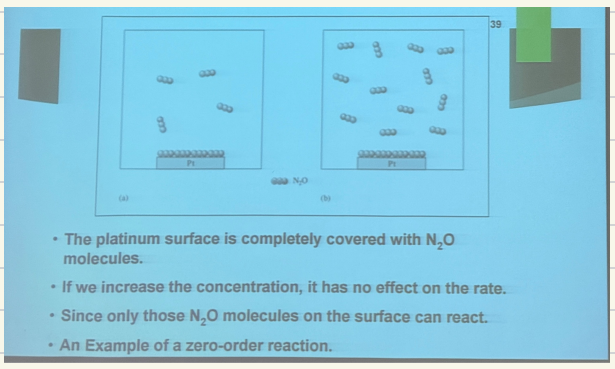


FIGURE 3.2 Variation in reactant concentration as a function of time for a zero-order reaction that consumes species A. The concentration of species A decreases linearly with time from an initial concentration c_{A_0}. The slope is given by the rate constant k.



First Order Reactions

Reaction rate is proportional to the amount of converted material

$$r = \frac{dx}{dt} = k_1(a-x)^1$$

If a - x = z, -dx = dz

$$\frac{dx}{a-x} = k_1 dt \qquad \frac{-dz}{z} = k_1 dt$$

$$-lnz = k_1 t + cst$$

$$-ln(a-x) = k_1 t + cst \quad t=0,\ x=0 \quad cst = -lna$$

$$-ln(a-x) = k_1 t - lna$$

$$ln\frac{a}{a-x} = k_1 t$$

$$k_1 = \frac{1}{t} ln\frac{a}{a-x}$$

$$k_1 = \frac{2.303}{t} log\frac{a}{a-x}$$

$$k_1 = \frac{2.303}{t} log\frac{C_0}{C}$$

$k_1 = sec^{-1}$

First Order Reactions

$$k_1 = \frac{1}{t} ln\frac{a}{a-x}$$

First order half life

$$x = \frac{a}{2}$$

$$t_{1/2} = \frac{1}{k_1} ln\frac{a}{a-\frac{a}{2}} = \frac{1}{k_1} ln2$$

$$t_{1/2} = \frac{0.693}{k_1}$$

Half-life is independent of the first concentration

Even if the initial concentration differs, the time required for the amount of substance to decrease by half is the same!!!

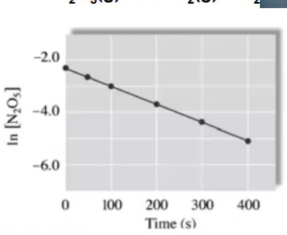
$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$

$[N_2O_5]$ (M)	Time (sec)
0.1000	0
0.0707	50
0.0500	100
0.0250	200
0.0125	300
0.00625	400

$\ln [N_2O_5]$	Time (sec)
-2.303	0
-2.649	50
-2.996	100
-3.689	200
-4.382	300
-5.075	400

1. Verify that the rate law is first order.
2. Calculate the rate constant.

The plot proves it is a straight line and so it is a first order reaction.



$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$

Since it is a first order reaction, we know that in the first order integrated rate law that k is equal to the -slope of the line.

$$Slope = \frac{\Delta Y}{\Delta X} = \frac{\Delta(\ln[N_2O_5])}{\Delta t}$$

$$Slope = \frac{-5.075-(-2.303)}{400-0} = \frac{-2.772}{400} = -6.93\times10^{-3}\ s^{-1}$$

$$k = -(slope) = 6.93\times10^{-3}\ s^{-1}$$

First Order Reactions

In a first-order reaction, the reaction rate is directly proportional to the concentration of one of the reactant concentrations. Thus, increasing the concentration of this reactant will speed up the rate of the reaction proportionally.

This behavior reflects the fundamental kinetic principle that the speed of most reactions depends on the frequency with which the reactants encounter one another.

Birinci dereceden bir reaksiyonda, reaksiyon hızı, reaktanlardan birinin konsantrasyonu ile doğru orantılıdır.

Bu reaktanın konsantrasyonunun artırılması, reaksiyon oranını orantılı olarak hızlandıracaktır.

$$\frac{dc_A}{dt} = -kc_A^1 = -kc_A$$

$$\frac{dc_A}{dt} = -kc_A$$

$$\int \frac{dc_A}{c_A} = -k\int dt$$

$$\ln \frac{c_A}{c_{A_0}} = -kt$$

$$c_A = c_{A_0} e^{-kt}$$

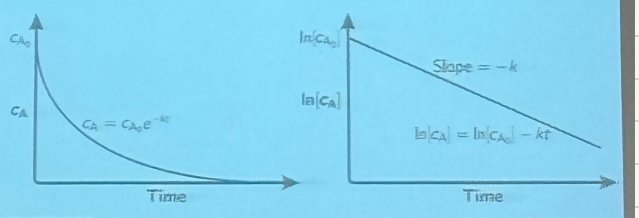


FIGURE 3.3 Variation in reactant concentration as a function of time for a first-order reaction that consumes species A. (a) The concentration of species A decreases exponentially with time from an initial concentration c_{A_0}. (b) A logarithmic plot of concentration versus time yields a straight line with a slope that is given by the rate constant k.

First Order Reactions

The half-life for the radioactive decay of carbon-14 is 5730 years. Calculate the rate constant for this decay reaction.

$$t_{1/2} = \frac{0.693}{k}$$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{5730\ \text{yr}}$$

$$k = 1.21\times10^{-4}/\text{yr} = 2.30\times10^{-10}/\text{min} = 3.83\times10^{-12}/\text{s}$$

Second Order Reactions

Reaction rate is proportional with the concentration squared

$$r = \frac{dx}{dt} = k_2(a-x)^2$$

$$\frac{dx}{(a-x)^2} = k_2 dt \quad \text{If a - x = z, -dx = dz}$$

$$\frac{-dz}{z^2} = k_2 dt$$

$$\int \frac{-dz}{z^2} = \int k_2 dt$$

$$\frac{1}{z} = k_2 t + cst$$

$$\frac{1}{a-x} = k_2 t + cst \quad t = 0,\ x = 0,\ cst = \frac{1}{a}$$

$$\frac{1}{a-x} = k_2 t + \frac{1}{a}$$

$$k_2 = \frac{1}{t}\frac{x}{a(a-x)}$$

$$k_2 = \frac{1}{t}\frac{C_0 - C}{C_0 * C}$$

Second Order Reactions

Second order half life

$$x = \frac{a}{2}$$

$$t = \frac{1}{k_2}\frac{x}{a(a-x)}$$

$$t_{1/2} = \frac{1}{k_2}\frac{\frac{a}{2}}{a(a-\frac{a}{2})}$$

$$t_{1/2} = \frac{1}{k_2 a}$$

Half-life is inversely proportional to the concentration

$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$

$[N_2O_5]$ (M)	Time (sec)
0.1000	0
0.0707	50
0.0500	100
0.0250	200
0.0125	300
0.00625	400

$\ln [N_2O_5]$	Time (sec)
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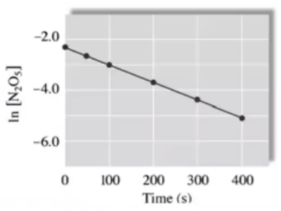

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$$Slope = \frac{\Delta Y}{\Delta X} = \frac{\Delta(\ln[N_2O_5])}{\Delta t}$$

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$$k = -(slope) = 6.93\times10^{-3}\ s^{-1}$$

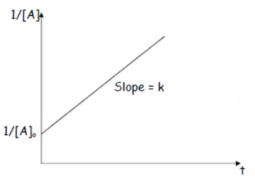
III) Second Order Reactions

a) Second order in one component

$A \rightarrow$ products

$$-\frac{d[A]}{dt} = k[A]^2$$

$$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$$



Third and nth Order Reactions

Third order

$$r = \frac{dx}{dt} = k_3(a-x)^3$$

$$k_3 = \frac{x(2a-x)}{2ta^2(a-x)^2}$$

$$t_{1/2} = \frac{3}{2k_3a^2}$$

nth order

$$r = \frac{dx}{dt} = k_n(a-x)^n$$

$$k_n = \frac{1}{(n-1)t}\left[\frac{1}{(a-x)^{n-1}} - \frac{1}{a^{n-1}}\right]$$

$$t_{1/2} = \frac{(2^{n-1}-1)}{(n-1)k_n a^{(n-1)}}$$

Summary of the Kinetics for Reactions of the Type aA → Products that are Zero, First, or Second Order i

	Order		
	Zero	*First*	
Rate law:	Rate = k	Rate = k[A]	Rate = k[A]^2
Integrated rate law:	[A] = $-kt$ + [A]$_0$	ln[A] = $-kt$ + ln[A]$_0$	$\frac{1}{[A]} = kt + \frac{1}{[A]_0}$
Plot needed to give a straight line:	[A] versus t	ln[A] versus t	$\frac{1}{[A]}$ versus t
Relationship of rate constant to the slope of straight line:	Slope = $-k$	Slope = $-k$	Slope = k
Half-life:	$t_{½} = \frac{[A]_0}{2k}$	$t_{½} = \frac{\ln 2}{k}$	$t_{½} = \frac{1}{k[A]_0}$

Key Equations

zeroth-order reaction

$\text{rate} = -\frac{\Delta[A]}{\Delta t} = k$

$[A] = [A]_0 - kt$

first-order reaction

$\text{rate} = -\frac{\Delta[A]}{\Delta t} = k[A]$

$[A] = [A]_0 e^{-kt}$

$\ln[A] = \ln[A]_0 - kt$

second-order reaction

$\text{rate} = -\frac{\Delta[A]}{\Delta t} = k[A]^2$

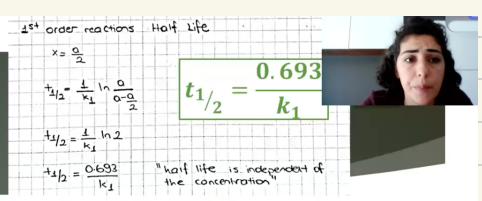
$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$

where [A]$_0$ i … at $t = 0$

Example

Degradation of uranium is a first order reaction and its half life is $t_{1/2} = 4.51x10^9$ years. Calculate the rate constant (k) and the time required (daily) for the 75% degration of uranium.

Example



Degradation of uranium is a first order reaction and its half life is $t_{1/2} = 4.51x10^9$ years. Calculate the rate constant (k) and the time required (daily) for the 75% degration of uranium.

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k} \qquad t = \frac{1}{k}\ln\frac{a}{a-x}$$

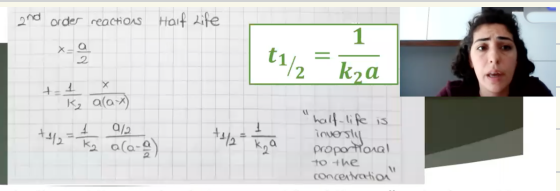
$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{4.51x10^9} \qquad t_{0.75} = \frac{1}{k}\ln\frac{a}{a-0.75a}$$

$$k = 1.532x10^{-10}\ 1/year \qquad = \frac{2.303x365}{1.532x10^{-10}} log4 = 330.4x10^{10} days$$

Example

In a second order reaction, initial concentration of the ractants is 0.4 mol/l. At the 80th minute, 30% of reaction is realized. Calculate the rate constant and the time required (in seconds) for the 80% of reaction realized.

Example



In a second order reaction, initial concentration of the ractants is 0.4 mol/l. At the 80th minute, 30% of reaction is realized. Calculate the rate constant and the time required (in seconds) for the 80% of reaction realized.

$$k = \frac{1}{t}\frac{x}{a(a-x)} \qquad k = \frac{1}{80x60}\frac{0.4x0.3}{0.4(0.4-0.4x0.3)} = 2.232x10^{-4}\ liter.second/mol$$

$$2.232x10^{-4} = \frac{1}{t_{0.8}}\frac{0.4x0.8}{0.4(0.4-0.4x0.8)}$$

$$t_{0.8} = \frac{0.4x0.8}{0.4(0.} = 4.48x10^4 seconds$$

Methods for Determining the Reaction Orders

- Integration Method
- Half-Life Method
- Van't Hoff Differential Method
- Linearized Coordinate / Graphic Method

Example

2nd order reactions Half Life

$$t_{1/2} = \frac{1}{k_2 a}$$

"half-life is inversly proportional to the concentration"

In a second order reaction, initial concentration of the ractants is 0.4 mol/l. At the 80th minute, 30% of reaction is realized. Calculate the rate constant and the time required (in seconds) for the 80% of reaction realized.

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Example

2nd order reactions Half Life

$$t_{1/2} = \frac{0.693}{k_1}$$

"half life is independent of the concentration"

Degradation of uranium is a first order reaction and its half life is $t_{1/2} = 4.51x10^9$ years. Calculate the rate constant (k) and the time required (daily) for the 75% degration of uranium.

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k} \qquad t = \frac{1}{k}\ln\frac{a}{a-x}$$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{4.51x10^9} \qquad t_{0.75} = \frac{1}{k}\ln\frac{a}{a-0.75a}$$

$$k = 1.532x10^{-10}\ 1/year \qquad = \frac{2.303x365}{1.532x10^{-10}} log4 = 330.4x10^{10} days$$

Second Order Reactions

$2NO_2(g) \rightarrow 2NO(g) + O_2(g)$

The decomposition of NO_2 to NO and O_2, has been experimentally determined to be second order with respect to the NO_2 concentration.
A reaction begins at time t = 0 with an unknown initial concentration of pure NO_2(g).
After 2 h, the rate of O_2 gas production is measured as 1.0 ×10−10 mol/(cm3·s).
If the reaction rate constant for this reaction is k = 10 $(mol/cm^3 \cdot s)^{-1}$, determine what the initial concentration of NO²(g) must have been when the reaction began.

$$\frac{dc_{NO_2}}{dt} = -k(c_{NO_2})^2$$

$$\frac{1}{c_{NO_2}} = \frac{1}{c_{NO_2 0}} + kt$$

$$\left.\frac{dc_{NO_2}}{dt}\right|_{t=2h} = -2\left.\frac{dc_{O_2}}{dt}\right|_{t=2h}$$

$$\left.\frac{dc_{NO_2}}{dt}\right|_{t=2h} = -2\left.\frac{dc_{O_2}}{dt}\right|_{t=2h} = -k(c_{NO_2}|_{t=2h})^2$$

$$c_{NO_2}|_{t=2h} = \sqrt{\left(\frac{2}{k}\right)\left.\frac{dc_{NO_2}}{dt}\right|_{t=2h}}$$

$$= \sqrt{\left(\frac{2}{10\ (mol/cm^3 \cdot s)^{-1}}\right) 1.00 \times 10^{-10}\ mol/(cm^3 \cdot s)}$$

$$= 4.47 \times 10^{-6}\ mol/cm^3$$

$$\frac{1}{c_{NO_2}|_{t=2h}} = \frac{1}{c_{NO_2 0}|_{t=2h}} + kt$$

$$c_{NO_2 0} = \frac{c_{NO_2}|_{t=2h}}{1 - c_{NO_2}|_{t=2h} kt}$$

$$= \frac{4.47 \times 10^{-6}\ mol/cm^3}{1 - 4.47 \times 10^{-6}\ mol/cm^3 \cdot 10\ (mol/cm^3 \cdot s)^{-1} \cdot 7200\ s} \times 10^{-6}\ mol/cm^3$$

Third and nth Order Reactions

Third order

$$r = \frac{dx}{dt} = k_3(a-x)^3$$

$$k_3 = \frac{x(2a-x)}{2ta^2(a-x)^2}$$

$$t_{1/2} = \frac{3}{2k_3 a^2}$$

nth order

$$r = \frac{dx}{dt} = k_n(a-x)^n$$

$$k_n = \frac{1}{(n-1)t}\left[\frac{1}{(a-x)^{n-1}} - \frac{1}{a^{n-1}}\right]$$

$$t_{1/2} = \frac{(2^{n-1}-1)}{(n-1)k_n a^{(n-1)}}$$

Summary of the Kinetics for Reactions of the Type aA → Products that are Zero, First, or Second Order i

	Order		
	Zero	*First*	
Rate law:	Rate = k	Rate = k[A]	Rate = k[A]²
Integrated rate law:	[A] = $-kt$ + [A]$_0$	ln[A] = $-kt$ + ln[A]$_0$	$\frac{1}{[A]} = kt + \frac{1}{[A]_0}$
Plot needed to give a straight line:	[A] versus t	ln[A] versus t	$\frac{1}{[A]}$ versus t
Relationship of rate constant to the slope of straight line:	Slope = $-k$	Slope = $-k$	Slope = k
Half-life:	$t_{1/2} = \frac{[A]_0}{2k}$	$t_{1/2} = \frac{\ln 2}{k}$	$t_{1/2} = \frac{1}{k[A]_0}$

Key Equations

zeroth-order reaction

$$rate = -\frac{\Delta[A]}{\Delta t} = k$$

$$[A] = [A]_0 - kt$$

first-order reaction

$$rate = -\frac{\Delta[A]}{\Delta t} = k[A]$$

$$[A] = [A]_0 e^{-kt}$$

$$\ln[A] = \ln[A]_0 - kt$$

second-order reaction

$$rate = -\frac{\Delta[A]}{\Delta t} = k[A]^2$$

$$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$$

TABLE 3.1 Summary of Zero-, First-, and Second-Order Reactions

	Zero Order	First Order	Second Order	
			c_A^2	$c_A c_B$
Rate law	$\frac{dc_A}{dt} = -k$	$\frac{dc_A}{dt} = -kc_A$	$\frac{dc_A}{dt} = -kc_A^2$	$\frac{dc_A}{dt} = -kc_A c_B$
Integrated	$c_A = c_{A_0} - kt$	$c_A = c_{A_0} e^{-kt}$	$\frac{1}{c_A} = \frac{1}{c_{A_0}} + kt$	$\frac{c_A}{c_B} = \frac{c_{A_0}}{c_{B_0}} \exp\left[kt\left(\frac{b}{a}c_{A_0} - c_{B_0}\right)\right]$
Units for k	$\frac{\text{concentration}}{\text{time}}$	$\frac{1}{\text{time}}$	$\frac{1}{\text{concentration} \cdot \text{time}}$	$\frac{1}{\text{concentration} \cdot \text{time}}$
Half-life	$\frac{c_{A_0}}{2k}$	$\frac{\ln 2}{k}$	$\frac{1}{kc_{A_0}}$	$-\frac{\ln\left(2 - \frac{b}{a}\frac{c_{A_0}}{c_{B_0}}\right)}{k\left(\frac{b}{a}c_{A_0} - c_{B_0}\right)}$

TABLE 3.1 Summary of Zero-, First-, and Second-Order Reactions

	Zero Order	First Order	Second Order	
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Units for k	$\frac{\text{concentration}}{\text{time}}$	$\frac{1}{\text{time}}$	$\frac{1}{\text{concentration} \cdot \text{time}}$	$\frac{1}{\text{concentration} \cdot \text{time}}$
Half-life	$\frac{c_{A_0}}{2k}$	$\frac{\ln 2}{k}$	$\frac{1}{kc_{A_0}}$	$-\frac{\ln\left(2 - \frac{b}{a}\frac{c_{A_0}}{c_{B_0}}\right)}{k\left(\frac{b}{a}c_{A_0} - c_{B_0}\right)}$

Integration Method – Reaction Rate

Differential equation that describes **the rate of change of a reactant (or product) concentration with time**. **If we integrate the rate law then we obtain an expression for the concentration as a function of time, which is generally the type of data obtained in an experiment**.

Reaction	Order	Differential form	Integrated form
$A \rightarrow P$	zeroth	$\frac{d[A]}{dt} = -k$	$[A] = [A]_0 - kt$
$A \rightarrow P$	first	$\frac{d[A]}{dt} = -k[A]$	$\ln[A] = \ln[A]_0 - kt$
$A + A \rightarrow P$	second	$\frac{1}{2}\frac{d[A]}{dt} = -k[A]^2$	$\frac{1}{[A]} = \frac{1}{[A]_0} + 2kt$
$A + B \rightarrow P$	second	$\frac{d[A]}{dt} = -k[A][B]$	$kt = \frac{1}{[B]_0 - [A]_0} \ln\frac{[B]_0[A]}{[A]_0[B]}$

In the above $[A]_0$ and $[B]_0$ represent the initial concentrations of A and B i.e. their concentrations at the start of the reaction.

If I plot the concentration, [A], as a function of time for each expression below, the correct order should yield a straight line function.

Zero order	First order	Second order
y = m x + b	y = m x + b	y = m x + b
[C] = -ck t + [C$_0$]	ln [C] = -ck t + ln [C$_0$]	1/[C] = ck t + 1/[C$_0$]

Methods for Determining the Reaction Orders

- Integration Method
- Linearized Coordinate / Graphic Method
- Half-Life Method
- Van't Hoff Differential Method

İntegrasyon Methodu

Deneysel olarak saptanan konsantrasyon ve zaman değerleri daha önce farklı reaksiyon dereceleri için saptanan eşitliklerde yerine koyularak, bu değerlerin (x ve t) hangi dereceden reaksiyona uygun olduğu araştırılarak reaksiyon derecesi belirlenir.

Reaksiyon derecesi: n
Başlangıç konsantrasyonu: C_0
t sürede reaksiyona giren kısım: x

$$-\frac{d(C_0 - x)}{dt} = \frac{dx}{dt} = k(C_0 - x)^n$$

Tablo I- Farklı Reaksiyon Dereceleri İçin Hız Eşitliklerinin Özeti

Derece	Hız Eşitliği: Diferansiyel denklem	Hız Eşitliği: İntegre edilmiş denklem	Hız Sabiti Birimi
0	$\frac{dx}{dt} = k$	$k = \frac{x}{t}$	mol/litre/sn
1/2	$\frac{dx}{dt} = k(C_0 - x)^{1/2}$	$k = \frac{2}{t}\left[C_0^{1/2} - x)^{1/2}\right]$	$mol^{1/2}/litre^{1/2}/sn$
1	$\frac{dx}{dt} = k(C_0 - x)$	$k = \frac{2.303}{t}\log\frac{C_0}{(C_0 - x)}$	1/ sn
3/2	$\frac{dx}{dt} = k(C_0 - x)^{3/2}$	$k = \frac{2}{t}\left[\frac{1}{(C_0 - x)^{1/2}} - \frac{1}{C_0^{1/2}}\right]$	$litre^{1/2}/mol^{1/2}/sn$
2	$\frac{dx}{dt} = k(C_0 - x)^2$	$k = \frac{1}{t}\left[\frac{x}{C_0(C_0 - x)}\right]$	litre/mol/sn
3	$\frac{dx}{dt} = k(C_0 - x)^3$	$k = \frac{1}{2t}\left[\frac{2C_0 x - x^2}{C_0^2(C_0 - x)^2}\right]$	$litre^2/mol^2/sn$

Linearized Coordinate / Graphic Method

In this method, coordinate systems are choosen to show the experimental data as a line in the graph.

Zeroth order	$x = k_0 t$	or	$C = C_0 - kt$
Fisrst oder	$ln\frac{a}{a - x} - t$	or	$lnC - t$
Second order			$\frac{1}{C} - t$
nth order			$\frac{1}{C^{n-1}} - t$

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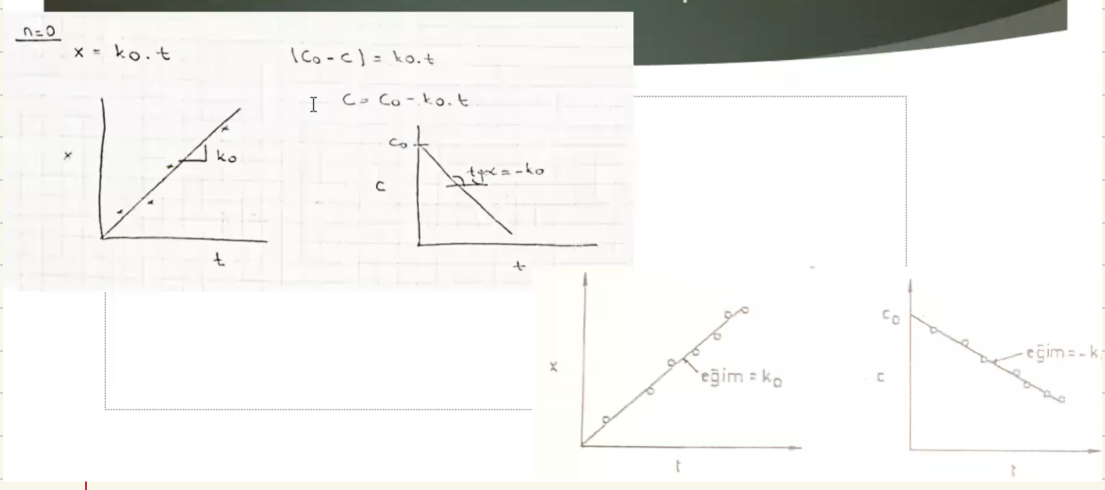
Linearized Coordinate / Graphic Method



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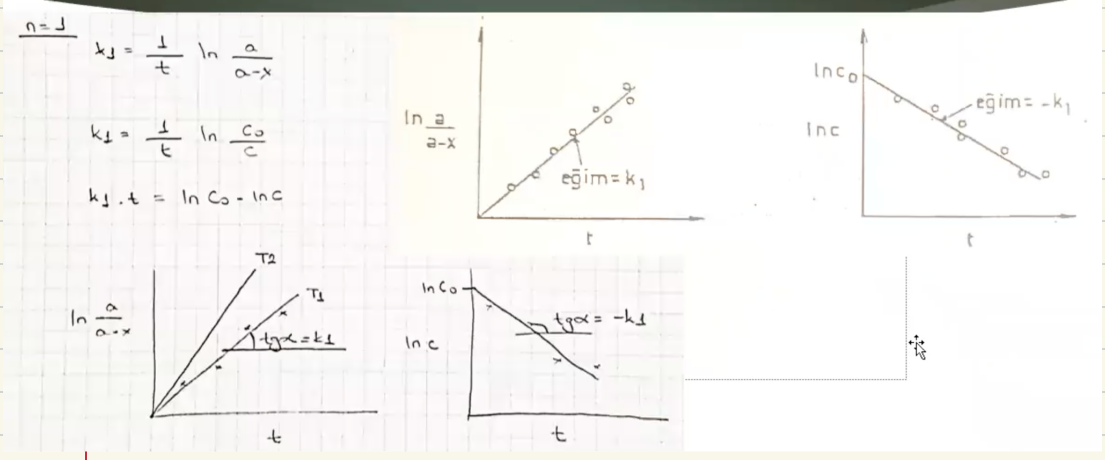
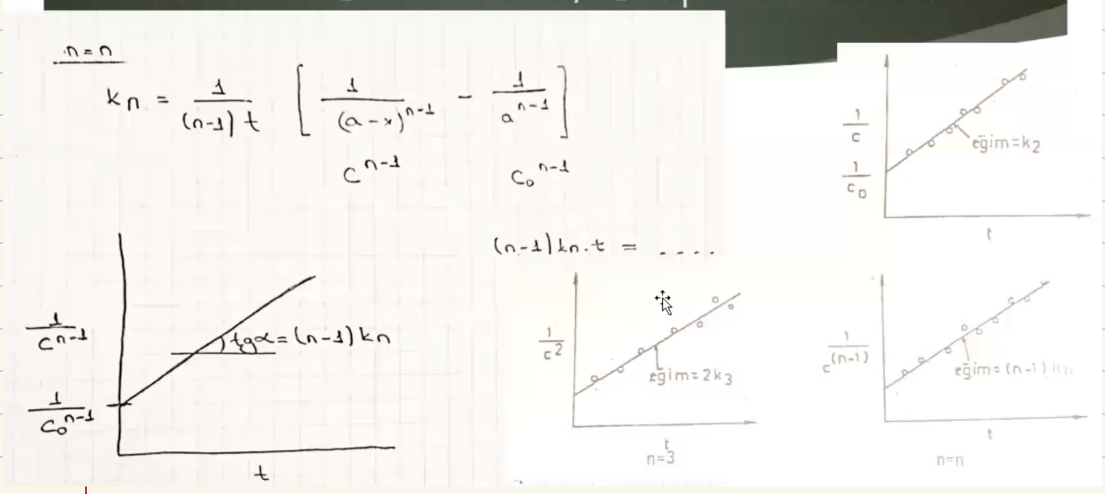


Linearized Coordinate / Graphic Method



Half Life Method – Reaction Rate

The half life, $t_{1/2}$, of a substance is defined as the time it takes for the concentration of the substance to fall to half of its initial value.

It only makes sense to define a half life for a substance not present in excess at the start of the reaction.

We can obtain equations for the half lives for reactions of various orders by substituting the values $t = t_{1/2}$ **and** $[A] = ½\,[A]_0$ (or a = a/2) into the integrated rate laws.

Zeroth order reaction	$t_{1/2} = \frac{[A]_0}{2k}$
First order reaction	$t_{1/2} = \frac{\ln 2}{k}$
Second order reaction	$t_{1/2} = \frac{1}{k[A]_0}$

Yarılanma Ömrü Yöntemi

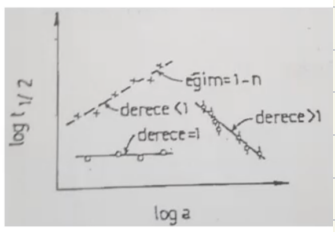
Bu yöntemde, reaktanların konsantrasyonları biribirine eşit ise, çeşitli reaksiyonlarda reaktanların başlangıç konsantrasyonlarının « **a** » yarıya düşmesi için gerekli yarılanma süreleri hesaplanır.

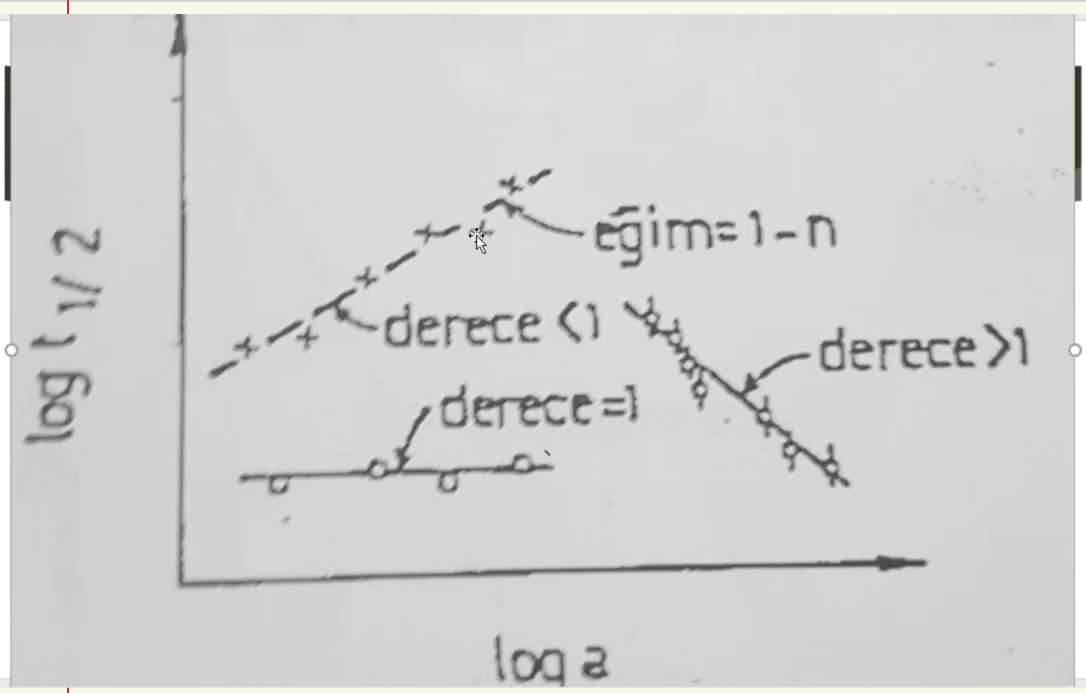
n=0	$t_{1/2} = \frac{a}{2k_0}$
n=1	$t_{1/2} = \frac{1}{k_1}$ $\ln 2 = \frac{0,693}{k_1}$
n=2	$t_{1/2} = \frac{1}{k_2 a}$
n=n	$t_{1/2} = \frac{2^{(n-1)}-1}{(n-1)ka^{(n-1)}}$

$$t_{1/2} = \frac{2^{n-1}-1}{(n-1)ka^{(n-1)}}$$

$$\log t_{1/2} = sabit - (n-1)\log a$$

Eğer reaksiyonun yarılanma ömrü farklı başlangıç konsantrasyonları için biliniyorsa grafik ya da hesaplama yöntemi ile reaksiyon derecesi hesaplanabilir.

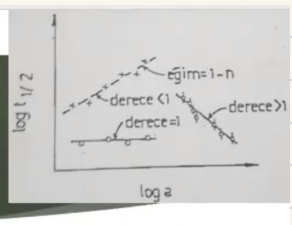




Example

$$\log t_{1/2} = cst - (n-1)\log a$$

$$t_{1/2} = \frac{1}{k_2 a}$$



For A+B=C reaction, calculate the reaction order and the rate constant.

Experiment	Initial Concentration		Half-life Time (h)
	A	B	
1	2	2	2.5
2	5	5	1.0

$$n = 1 + \frac{\log t_{1/2\,1} - \log t_{1/2\,2}}{\log a_2 - \log a_1} \qquad n = 1 + \frac{\log 2.5 - \log 1}{\log 5 - \log 2} \qquad n = 1 + \frac{\log 2.5}{\log 2.5} = 2$$

$$k = \frac{1}{a \times t_{1/2}} = \frac{1}{2 \times 2.5} = 0.2 \text{ liter/mol/hour}$$

Example

For A+B=C reaction, calculate the reaction order and the rate constant.

Experiment	Initial Concentration		Half-life Time (h)
	A	B	
1	2	2	2.5
2	5	5	1.0

Example

For a calcinaton reaction, calculate the reaction order and rate constant.

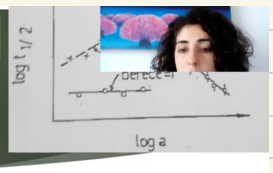
Initial Concentration (mol/l)	0.5	1.1	2.48
Half-lite time (sec)	4280	885	174

log a	-0.3010	0.0414	0.3945
log $t_{1/2}$	3.6314	2.9464	2.2405

Example

$$log t_{1/_2} = cst - (n-1)\log a$$

$$t_{1/_2} = \frac{(2^{n-1}-1)}{(n-1)k_n a^{(n-1)}}$$



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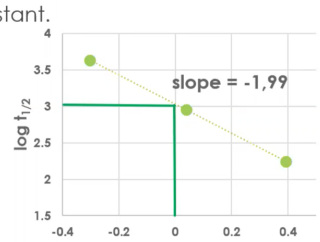


$$slope = -(n-1) = -1.99$$

$$n = 2.99 \cong 3$$

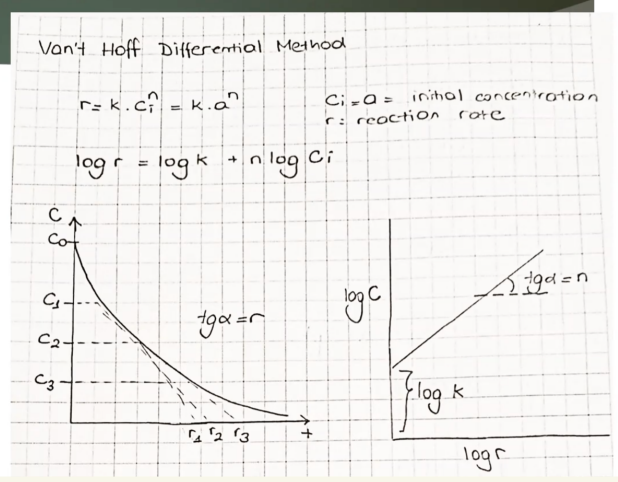
for log a = 0

$$log t_{1/_2} = log \frac{2^{3-1}-1}{(3-1)k} = log \frac{3}{2k_3} = 3.03$$

$$k_3 = 1.399 \times 10^{-3} \text{ liter}^2/\text{mol}^2/\text{sec}$$

Van't Hoff Differential Method

In this method one utilizes no concentrations at given time but rates of reactions at given time.
It may be employed when reactions are not very fast. Most frequently it is utilized for initial rates.

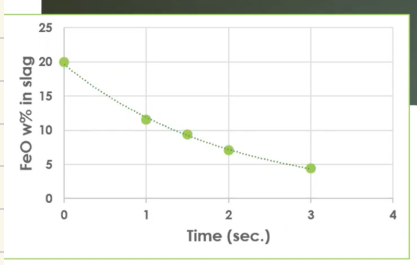


Example

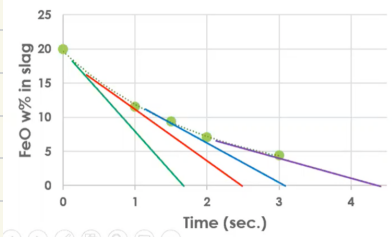
The reduction kinetics at a certain temperature between the C in liquid iron and FeO in slag is observed and obtained data is given below.

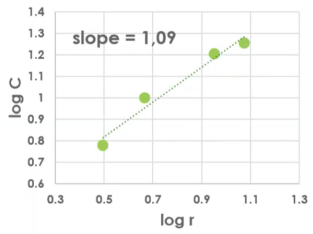
According to the table, determine the order of reaction by using Van't Hoff differential method and calculate the reaction rate constant (k).

FeO in slag(w%)	20.00	11.50	9.35	7.10	4.40
Time (second)	0	1.0	1.5	2.0	3.0



Tangent #	FeO in slag (w%) - C	log C	Slope of tangent	Reduction rate (r)	log r
1	18	1.2553	-11.88	11.88	1.0749
2	16	1.2041	-8.94	8.94	0.9513
3	10	1.000	-4.63	4.63	0.6656
4	6	0.7782	-3.13	3.13	0.4955





$$r = kC^n$$

$$\log r = \log k + n \log C$$

$$\text{at } \log C = 0$$

$$\log k = -0{,}34$$

$$k = 0{,}4571 \text{ l/min}$$